

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	17 July 2026
Team ID	-----
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement:

Develop an intelligent flood prediction system using machine learning to predict flood risk from rainfall and meteorological data, enabling early warning and disaster preparedness.

Step-2: Brainstorm, Idea Listing and Grouping

S.No	Team Member	Idea / Suggestion	Category	Group No.
1.	Member 1	Collect an open-source flood prediction dataset containing annual rainfall, seasonal rainfall, cloud visibility and meteorological parameters	Data Collection	1

		from Kaggle for model development		
2.	Member 2	Perform Exploratory Data Analysis (EDA) using distribution plots, box plots, heat maps and descriptive statistics to understand rainfall patterns and identify important features affecting floods	Data Analysis	2
3.	Member 3	Preprocess the dataset by handling missing values, treating outliers using IQR capping, encoding categorical features, splitting training/testing data and applying feature scaling.	Data Preprocessing	3
4	Member 4	Train and	Machine	4

		compare Decision Tree, Random Forest, KNN and XGBoost models to identify the best-performing flood prediction algorithm based on accuracy and evaluation metrics. Deploy the selected XGBoost model using Flask to provide an interactive web application where users can enter weather parameters and instantly predict flood risk.	Learning. Web Deployment	
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Step-3: Idea Prioritization

Group NO	Final Idea	Feasibility	Importance	Prority	Selected
1	Collect and prepare a	High	High	5	No

	reliable flood prediction dataset				
2	Perform EDA to identify important rainfall and weather features	High	High	4	No
3	Preprocess the dataset for accurate model training	High	High	3	No
4	Develop and compare multiple ML models (Decision Tree, Random Forest, KNN and XGBoost)	High	High	2	No
5	Deploy the best-performing XGBoost model as a Flask-based Flood Prediction System	High	High	1	Yes